# Ch 3.1: Linear Regression

## Lecture 4 - CMSE 381

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# Announcements

Last time:

- 2.2 Assessing Model Accuracy
- Test your understand: PollEv

# Covered in this lecture

- Least squares coefficient estimates for linear regression
- Residual sum of squares (RSS)

Section 1

# Simple Linear Regression

# Setup

- Predict $Y$ on a single predictor variable $X$

$$Y \approx \beta_0 + \beta_1 X$$

- "$\approx$" .... "is approximately modeled as"

# Example

| | TV | Radio | Newspaper | Sales |
|---|---|---|---|---|
| 1 | 230.1 | 37.8 | 69.2 | 22.1 |
| 2 | 44.5 | 39.3 | 45.1 | 10.4 |
| 3 | 17.2 | 45.9 | 69.3 | 9.3 |
| 4 | 151.5 | 41.3 | 58.5 | 18.5 |
| 5 | 180.8 | 10.8 | 58.4 | 12.9 |
| 6 | 8.7 | 48.9 | 75 | 7.2 |
| 7 | 57.5 | 32.8 | 23.5 | 11.8 |
| 8 | 120.2 | 19.6 | 11.6 | 13.2 |
| 9 | 8.6 | 2.1 | 1 | 4.8 |
| 10 | 199.8 | 2.6 | 21.2 | 10.6 |
| 11 | 66.1 | 5.8 | 24.2 | 8.6 |

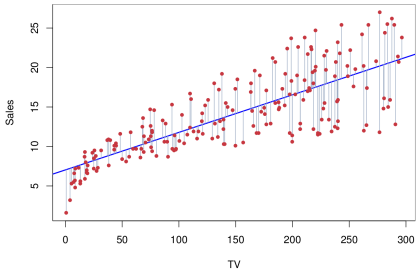
- $$\text{sales} \approx \beta_0 + \beta_1 \text{TV}$$
- $\beta_0$ intercept; $\beta_1$ slope
- Coefficients or parameters : $\{\beta_0, \beta_1\}$
- Once we have good guesses for $\hat{\beta}_i$, model is

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

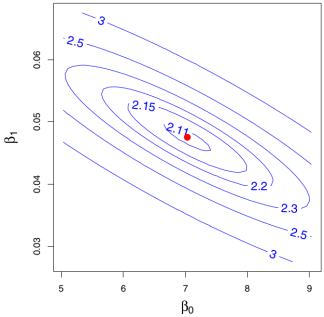
# Least squares criterion: Setup

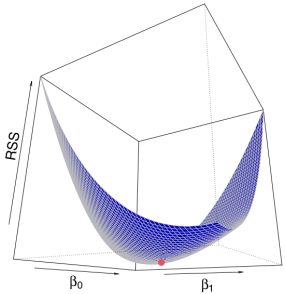
## How do we estimate the coefficients?

- Given $(x_1, y_1), \cdots, (x_n, y_n)$
- Let $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ be prediction for $Y$ on $i$th value of $X$.
- $e_i = y_i - \hat{y}_i$ is the $i$th residual
-



# Least squares criterion: RSS




Residual sum of squares RSS is

$$RSS = e_1^2 + \cdots + e_n^2$$
$$= \sum_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

$$\text{sales} \approx \beta_0 + \beta_1 \text{TV}$$

**Least squares criterion**

Find $\beta_0$ and $\beta_1$ that minimize the RSS.

# Least squares coefficient estimates

*Minimizing RSS:*

$$\min_{\beta_0,\beta_1} \sum_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

*There's a typo on old versions textbook?*

$$\frac{\partial RSS}{\partial \hat{\beta}_0} = -2 \sum_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

$$\frac{\partial RSS}{\partial \hat{\beta}_1} = -2 \sum_i x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \overline{x})(y_i - \overline{y})}{\sum_{i=1}^n (x_i - \overline{x})^2}$$

$$\hat{\beta}_0 = \overline{y} - \hat{\beta}_1 \overline{x}$$

- Closed form!
- Though iterative methods (e.g. gradient descent) is often used for larger problems

# Coding group work

# Next time

Next time:

- More on simple linear regression!
- Evaluation of model etc.

CMSE381_F2025_Schedule : Schedule

| Lec # | Date | | Topic | Reading | HW |
|---|---|---|---|---|---|
| 1 | M | 8/25 | Intro / Python Review | 1 | |
| 2 | W | 8/27 | What is statistical learning | 2.1 | |
| 3 | F | 8.29 | Assessing Model Accuracy | 2.2.1, 2.2.2 | |
| | M | 9/1 | Labor Day - No Class | | |
| 4 | W | 9/3 | Linear Regression | 3.1 | |
| 5 | F | 9/5 | More Linear Regression | 3.1 | HW #1 Due Sun 9/7 |
| 6 | M | 9/8 | Multi-linear Regression | 3.2 | |
| 7 | W | 9/10 | Probably More Linear Regression | 3.3 | |
| 8 | F | 9/12 | Last of the Linear Regression | | |

**Announcements**

- Homework 1
  - Due Sun, Sep 7